

ESE 218 – Project 3

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I. INTRODUCTION

The goal of this project was to explore how Principal Component Analysis (PCA) can be used for image compression and low-dimensional feature extraction. I worked with the Extended Yale B face dataset, which contains 15 subjects with many images each taken under different lighting conditions and facial expressions. Because each image has thousands of pixels, the dataset is extremely high-dimensional, making storage and pattern extraction difficult.

In this project, I used Singular Value Decomposition (SVD) to perform PCA on the training images and identify a set of low-dimensional basis vectors (principal components) that capture most of the variation in the dataset. I then used these components to approximate, compress, and reconstruct the training images, as well as evaluating how well the same PCA features generalize to test images and a rotated face. Overall, the project demonstrates how high-dimensional data can be represented using only a small number of principal components.

II. BACKGROUND

PCA provides a way to understand and reduce the dimensionality of high-dimensional datasets such as images. When an image with M pixels is reshaped into a vector in R^M , a dataset of N images forms a matrix

$$X \in R^{M \times N},$$

where each column is one vectorized image. Since raw pixel values are usually highly correlated, most of the meaningful variation in the data can be represented using a much smaller number of components.

PCA assumes the data are centered so that variations are measured relative to the average image. PCA is computed using the Singular Value Decomposition (SVD), which factorizes the data matrix as

$$X = USV^T.$$

The decomposition has several important properties for component analysis:

- The columns of U are orthonormal vectors in R^M and represent the principal components (or eigenfaces). Each of the columns corresponds to a direction of maximum variation in the dataset.
- The diagonal entries of S are the singular values, and their squared magnitudes indicate how much variance each principal component captures.

- The rows of V^T contain the coefficients or feature values expressing each image in the PCA basis.

By keeping only the first k principal components, we get the best rank- k approximation of the dataset with the least-squares:

$$\hat{X}_k = U_k S_k V_k^T.$$

This provides a method for image compression. Instead of storing all M pixels, an image can be represented using only k PCA coefficients. When $k \ll M$, there is significantly less storage used while still keeping the key structure of the images.

III. METHODS AND RESULTS

CONSTRUCTION OF THE DATA MATRIX AND APPROXIMATION ANALYSIS

A. Forming the data matrix

Each image in the Extended Yale B dataset is stored as a grayscale array. For every subject $s = 1, \dots, 13$, all images were read, converted to double precision, in order to include decimals, and reshaped into column vectors

$$x_i \in R^M,$$

where M is the total number of pixels. Stacking the vectors produced the training data matrix

$$X = [x_1 \ x_2 \ \dots \ x_N] \in R^{M \times N}.$$

In total, there were $N = 143$ training images loaded, giving a matrix $X \in R^{11368 \times 143}$.

B. Singular Value Decomposition and Variance

To analyze the variation in the dataset, the Singular Value Decomposition (SVD) was computed:

$$X = USV^T.$$

The diagonal entries σ_i of the matrix S are the singular values. Their squared magnitudes measure the amount of variance in the i th principal component:

$$\text{variance}_i = \sigma_i^2.$$

The total data energy is

$$\|X\|_F^2 = \sum_{i=1}^r \sigma_i^2, \quad r = \text{rank}(X).$$

Components with large singular values include the most important features of the images, while components with very

small singular values come from noise or minor fluctuations. The cumulative sum of the squared singular values

$$E_k = \sum_{i=1}^k \sigma_i^2$$

indicates how much of the dataset's energy is kept when only the first k components are used. This is a direct way to quantify how well a rank- k approximation represents the training images and determines how many components are needed to achieve a desired accuracy. As shown in Figure 1, the singular values decay rapidly, confirming that most of the data's variance is concentrated in the first few components.

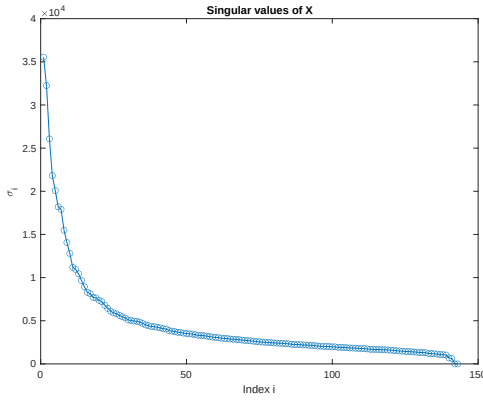


Fig. 1. Singular values of the data matrix X . The rapid decay indicates that most variation is captured by the first few principal components.

C. Error Thresholds and Compression Factor

The reconstruction error of the best rank- k approximation $\hat{X}_k = U_k S_k V_k^T$ is

$$\frac{\|X - \hat{X}_k\|_F^2}{\|X\|_F^2} = \frac{\sum_{i=k+1}^r \sigma_i^2}{\sum_{i=1}^r \sigma_i^2}.$$

Using this expression, the number of principal components required to achieve various accuracy levels was computed:

- 30% error: $k = 10$
- 20% error: $k = 16$
- 10% error: $k = 34$
- 5% error: $k = 59$

This shows that only a very small number of components capture most of the variation in the training set. As seen in Figure 2, the Frobenius reconstruction error decreases rapidly

with increasing k , reflecting the fast decay of the singular values.

Storing an image directly requires all $M = 11368$ pixel values. To recover 90% of the image energy (10% error), only $k = 34$ principal components are needed. The compression factor is

$$\frac{11368}{34} \approx 334.35.$$

This means PCA compresses each image by a factor of over 300 while still retaining most of the meaningful structure.

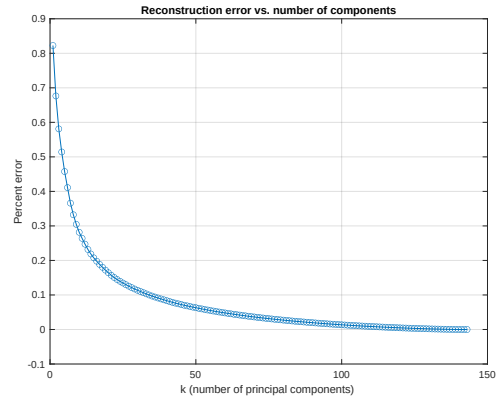


Fig. 2. Frobenius reconstruction error $\|X - \hat{X}_k\|_F^2 / \|X\|_F^2$ as a function of the number of principal components k .

D. Approximation Quality

Rank- d reconstructions were computed for $d = 20, 50, 70, 100$. For each value of d , the Frobenius percent error was computed using

$$\text{FrobError}(d) = \frac{\|X - \hat{X}_d\|_F^2}{\|X\|_F^2},$$

where $\hat{X}_d = U_d S_d V_d^T$ is the best rank- d approximation of X . The resulting Frobenius percent-errors were:

- $d = 20$: 16.44%
- $d = 50$: 6.30%
- $d = 70$: 3.63%
- $d = 100$: 1.36%

To find image-by-image accuracy, the average per-image relative error was computed using

$$\text{RelError}_j(d) = \frac{\|x_j - \hat{x}_{j,d}\|_2^2}{\|x_j\|_2^2},$$

$$\text{AvgRelError}(d) = \frac{1}{N} \sum_{j=1}^N \text{RelError}_j(d),$$

where x_j is the j th original image and $\hat{x}_{j,d}$ is its reconstruction from d components. The average per-image relative errors were:

- $d = 20$: 90.09%
- $d = 50$: 88.29%
- $d = 70$: 87.82%
- $d = 100$: 87.42%

The Frobenius percent error decreases significantly as d increases, showing that the leading principal components capture most of the total dataset variation. However, the per-image relative errors is high because the metric is more sensitive to fine pixel-level details that are not captured unless a much larger number of components is used. PCA provides an great low-dimensional representation of the overall dataset structure, even though individual images still differ substantially from their low-rank approximations.

E. Visual reconstructions

One training image was reconstructed using $d = 20, 50, 70$, and 100 components. As d increases, more structure and detail are restored. Lower values of d produce smooth “eigenface-like” images reflecting only the dominant lighting and shape variations, while higher values capture more facial detail. As shown in Figure 3, the improvement in visual quality matches what is expected from the singular value decay.

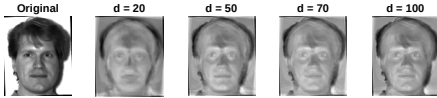


Fig. 3. Reconstruction of a training image using $d = 20, 50, 70$, and 100 principal components. Higher d values restore more detail.

USING PCA FEATURES TO COMPRESS THE TEST DATASET

After computing PCA using the training images (subjects 1-13), the next step was to evaluate how well the PCA basis generalizes to new images. The images from subjects 14 and 15 were loaded and vectorized in the same way as the training set. These images were then centered using the training-set mean, and their PCA feature representations were found by projecting onto the first $d = 50$ principal components:

$$Z_{\text{test}} = U_{50}^{\top} (X_{\text{test}} - \mu).$$

Each test image is therefore represented using only 50 coefficients instead of all 11368 pixels.

To measure reconstruction quality, the test images were reconstructed from their 50-dimensional PCA features:

$$\hat{X}_{\text{test}} = \mu + U_{50} Z_{\text{test}}.$$

The Frobenius reconstruction error for the entire test set was

$$\frac{\|X_{\text{test}} - \hat{X}_{\text{test}}\|_F^2}{\|X_{\text{test}}\|_F^2} = 0.0434 \quad (4.34\%).$$

This relatively small error shows that the PCA features extracted from the training subjects transfer well to unseen subjects, meaning the dominant variance learned through PCA, overall face shape and large illumination patterns, are shared across all individuals in the dataset.

Rotated Test Image

To test PCA’s adaptability to geometric changes, the image `subject15.normal.pgm` was rotated by 25° and projected onto the same 50-dimensional PCA space. The reconstruction followed the same formula:

$$\hat{x}_{\text{rot}} = \mu + U_{50} U_{50}^{\top} (x_{\text{rot}} - \mu).$$

The relative error for this rotated image was significantly higher:

$$\frac{\|x_{\text{rot}} - \hat{x}_{\text{rot}}\|_2^2}{\|x_{\text{rot}}\|_2^2} = 0.2665 \quad (26.65\%).$$

The large error reflects an important limitation of PCA: because PCA is a linear method trained on unrotated faces, it cannot represent geometric transformations like rotation using the same components. While PCA is able to capture global lighting and structural variation, it cannot account for nonlinear changes that were not in the training data. As shown in Figure 4, the reconstruction loses significant structural detail.

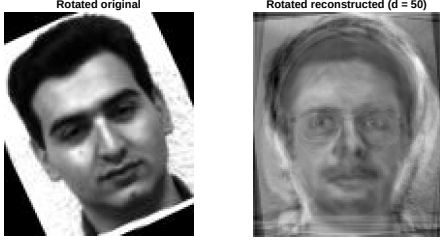


Fig. 4. Reconstruction of the rotated test image using $d = 50$ principal components. PCA struggles with geometric transformations such as rotation.

Overall, the test results demonstrate that PCA provides strong compression performance on new images with similar structure, but its effectiveness drops significantly when the input has transformations not in the learned subspace from the training set.

IV. DISCUSSION AND CONCLUSION

This project used PCA and the SVD to analyze and compress facial images from the Extended Yale B dataset. By stacking all training images into a data matrix and applying SVD, a small set of principal components was identified that captured most of the variation across subjects and lighting conditions. The rapid decay of the singular values showed that the first few components contain almost all of the meaningful structure. For example, using only 34 components preserved 90% of the dataset energy, reducing each image from 11,368 pixels to just 34 numbers.

The approximation results highlighted a difference between global and per-image accuracy. The Frobenius percent error dropped quickly with increasing d , reaching only 1.36% error at $d=100$, showing that the leading principal components provide a good low-rank model of the entire dataset. However, the average per-image relative error remained high (about 88–90%), mostly because the Yale B images vary heavily in lighting. Even small shadow differences create large pixel-level mismatches, so PCA captures the shared structure of the dataset but not the fine details of each individual image.

The test set experiments showed similar behavior. Projecting the images from subjects 14 and 15 into the 50-dimensional

PCA space resulted in only 4.34% reconstruction error, meaning the PCA basis generalizes well to new subjects. However, a rotated face produced a much higher error (26.65%), reflecting PCA's limitation with geometric transformations that were not present in the training set.

Overall, PCA proved to be a useful method for compressing and analyzing high-dimensional images, allowing for large reductions in storage while preserving the dominant facial structure. Its main limitations come from its linearity, which makes it less effective at handling nonlinear variations such as lighting extremes, expression changes, and rotations. Future work could explore nonlinear approaches, such as kernel PCA, that can capture these more complex transformations.

V. REFERENCES

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- [2] B. Schölkopf, A. Smola, and K.-R. Müller, "Kernel principal component analysis," *Artificial Neural Networks — ICANN 1997*, Berlin, Heidelberg: Springer, pp. 583–588, 1997. Available: https://graphics.stanford.edu/courses/cs233-25-spring/ReferencedPapers/scholkopf_kernel.pdf